FIN2020 HW4 文杰 123090612

1. The Gains From Diversification

Using the two-asset portfolio formulas from MPT, the portfolio's expected return is

$$\mu_P = w\mu_1 + (1-w)\mu_2$$

and its standard deviation is

$$\sigma_P = \left[w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\sigma_1\sigma_2\rho_{12}\right]^{1/2}.$$

Given:

$$\mu_1 = 10, \quad \sigma_1 = 8, \quad \mu_2 = 6, \quad \sigma_2 = 6, \quad w = \frac{1}{2}.$$

a. When $\rho_{12} = 0$

$$\mu_P = \frac{1}{2}(10) + \frac{1}{2}(6) = 5 + 3 = 8.$$

$$\sigma_P = \left[\left(\frac{1}{2}\right)^2 8^2 + \left(\frac{1}{2}\right)^2 6^2 + 2\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)(8)(6)(0)\right]^{1/2} = 5$$

b. When $\rho_{12} = -0.50$

$$\mu_P = \frac{1}{2}(10) + \frac{1}{2}(6) = 8.$$

$$\sigma_P = \left[\left(\frac{1}{2}\right)^2 8^2 + \left(\frac{1}{2}\right)^2 6^2 + 2\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)(8)(6)(-0.50)\right]^{1/2} = \sqrt{13}$$

c. For which value of ρ_{12} are the gains from diversification larger?

The expected return is the same in both cases, but the portfolio risk is lower when correlation is more negative, so the gains from diversification are larger when $\rho_{12} = -0.50$.

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2. Portfolio Allocation with Mean-Variance Utility

a. Expected return

Let the portfolio return be $\tilde{r}_P$. Since the investor allocates fraction w to the risky asset and fraction $1-w$ to the risk-free asset, the portfolio return is

$$\tilde{r}_P = w\tilde{r} + (1-w)r_f.$$

Take expectations of it:

$$\mu_P = E(\tilde{r}_P) = w\mu_r + (1-w)r_f.$$

b. Variance of return

Because r_f is non-random, it has zero variance and zero covariance with the risky part. Hence

$$\sigma_P^2 = \mathrm{Var}(\tilde{r}_P) = \mathrm{Var}(w\tilde{r} + (1-w)r_f) = w^2\sigma_r^2.$$

c. Optimal choice w^*

Substitute the expressions from parts (a) and (b) into the utility function:

$$U(\mu_P, \sigma_P) = \mu_P - \left(\frac{A}{2}\right)\sigma_P^2$$

so

$$U(w) = \left[w\mu_r + (1-w)r_f\right] - \frac{A}{2}w^2\sigma_r^2.$$

Rearrange:

$$U(w) = r_f + w(\mu_r - r_f) - \frac{A}{2}w^2\sigma_r^2.$$

Now maximize with respect to w. The first-order condition is

$$\frac{dU}{dw} = (\mu_r - r_f) - Aw\sigma_r^2 = 0.$$

Solving gives

$$\boxed{w^* = \frac{\mu_r - r_f}{A\sigma_r^2}}.$$

Since the utility is quadratic in w with second derivative

$$\frac{d^2U}{dw^2} = -A\sigma_r^2 < 0$$

for $A > 0$, this is indeed the optimal choice. This matches the mean-variance portfolio logic in the notes: the risky share rises with the excess expected return $(\mu_r - r_f)$ and falls with both risk aversion A and risk σ_r^2.

The final answers are:

$$\boxed{\mu_P = w\mu_r + (1-w)r_f}$$

$$\boxed{\sigma_P^2 = w^2\sigma_r^2}$$

$$\boxed{w^* = \frac{\mu_r - r_f}{A\sigma_r^2}}.$$